- D'Alembert Principle of Virtual Work

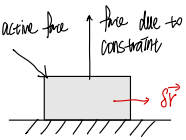
$$F_i = m_i \frac{d}{dt}\vec{v}_{i0} \qquad i=1,\cdots,N$$

$$\Downarrow$$

$$\underbrace{F_i}_{\text{actual force}} - m_i \frac{d}{dt}\vec{v}_{i0} = 0 \qquad i=1,\cdots,N$$

↳ $F_i = F_i^{(act)} + F_i^{(cons)}$ = active force + force due to constraints

Eg.



$\delta\vec{r}$ is an admissible "virtual displacement"

"admissible" means satisfying the constraints

Virtual work: virtual work is the work done by the force moving a particle through an admissible virtual displacement

$$\delta W = \langle \vec{F}, \delta\vec{r} \rangle$$

$$= \langle \vec{F}^{(act)}, \delta\vec{r} \rangle + \underbrace{\langle \vec{F}^{(cons)}, \delta\vec{r} \rangle}_{\delta W^{(cons)} = 0}$$

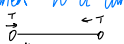
$$= \langle \vec{F}^{(act)}, \delta\vec{r} \rangle$$

the constraint force do no work

the constraints are called "workless constraints"

Some workless constraints

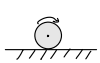
- mass constrained to a curve / surface
- rigidity



- rope-pulley system



- rolling without slipping



- D'Alembert principle of virtual work

let q be the generalized coordinate

$r_i(q) \Rightarrow \delta r_i = Dr_i(q)\,\underbrace{\delta\vec{q}}_{\text{admissible virtual displacement of generalized coordinates}}$

$$F_i - m_i \frac{d}{dt}\vec{v}_{i0} = 0$$

$$F_i^{(act)} + F_i^{(cons)} - m_i \vec{a}_{i0} = 0$$

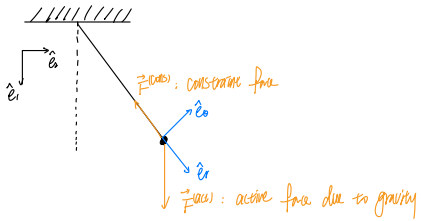
$$\langle F_i^{(act)}, \delta r_i \rangle + \langle F_i^{(cons)}, \delta r_i \rangle - m_i \langle \vec{a}_{i0}, \delta r_i \rangle = 0$$

$$\sum_{i=1}^{m} \langle F_i^{(act)} - m_i \vec{a}_{i0}, \delta r_i \rangle + \underbrace{\sum_{i=1}^{m} \langle F_i^{(cons)}, \delta r_i \rangle}_{=0 \text{ think above rigidity}} = 0$$

$$\sum_{i=1}^{m} \langle F_i^{(act)} - m_i \vec{a}_{i0}, \delta r_i \rangle = 0$$

D'Alembert principle of virtual work

Eg. pendulum



1 DOF: free to move in θ

$q = \theta$ is the generalized coordinate

$\delta \vec{r} = l\,\delta\theta\,\hat{e}_\theta$

$\vec{a}_{p/o} = l\ddot{\theta}\hat{e}_\theta - l\dot{\theta}^2\hat{e}_r$

$\langle \vec{F}^{(act)} - m\vec{a}_{p/o},\ \delta\vec{r}_{p/o} \rangle = 0$

$\langle mg\hat{e}_1 - ml\ddot{\theta}\hat{e}_\theta + ml\dot{\theta}^2\hat{e}_r,\ l\,\delta\theta\,\hat{e}_\theta \rangle = 0$

$(mg\,\hat{e}_1 \cdot \hat{e}_\theta - ml\ddot{\theta})\delta\theta = 0 \quad \forall \delta\theta$

$mg\sin\theta - ml\ddot{\theta} = 0$

- D'Alembert Principle in Minimal Set of Generalized Coordinate

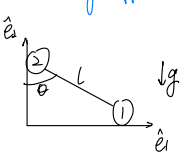
$\sum_{i=1}^{N} \langle \vec{F}_i^{(act)} - m_i\vec{a}_{i/o},\ \delta\vec{r}_i \rangle = 0$

$\sum_{i=1}^{N} \langle \vec{F}_i^{(act)} - m_i\vec{a}_{i/o},\ \frac{\partial \vec{r}_i}{\partial q}\delta q \rangle = 0$ $\quad \delta q$ is free

$\sum_{i=1}^{N} \left(\vec{F}_i^{(act)} - m_i\vec{a}_{i/o}\right)^T \frac{\partial \vec{r}_i}{\partial q} = 0$

turns into 2nd order ODEs in q

Eg. Baton sliding off the wall (without friction)



1 DOF, $q = \theta$

$\vec{r}_{1/o} = l\sin\theta\,\hat{e}_1$ $\quad \vec{v}_{1/o} = l\cos\theta\dot{\theta}\hat{e}_1$ $\quad \vec{a}_{1/o} = l(\cos\theta\ddot{\theta} - \sin\theta\dot{\theta}^2)\hat{e}_1$

$\vec{r}_{2/o} = l\cos\theta\,\hat{e}_2$ $\quad \vec{v}_{2/o} = -l\sin\theta\dot{\theta}\,\hat{e}_2$ $\quad \vec{a}_{2/o} = l(-\sin\theta\ddot{\theta} - \cos\theta\dot{\theta}^2)\hat{e}_2$

$\frac{\partial \vec{r}_{1/o}}{\partial q} = l\cos\theta\,\hat{e}_1$ $\quad \frac{\partial \vec{r}_{2/o}}{\partial q} = -l\sin\theta\,\hat{e}_2$

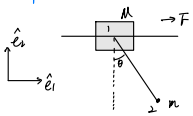
D'Alembert principle: $\sum_{i=1}^{N} \left(\vec{F}_i^{(act)} - m_i\vec{a}_{i/o}\right)^T \frac{\partial \vec{r}_{i/o}}{\partial q} = 0$

$$\left[-mg\hat{e}_2 - ml(\cos\theta\ddot{\theta} - \sin\theta\dot{\theta}^2)\hat{e}_1\right]^T l\cos\theta\hat{e}_1$$
$$+ \left[-mg\hat{e}_2 + ml(\sin\theta\ddot{\theta} + \cos\theta\dot{\theta}^2)\hat{e}_2\right]^T \cdot -l\sin\theta\hat{e}_2 = 0$$

$$-ml^2(\cos\theta\ddot{\theta} - \sin\theta\dot{\theta}^2)\cos\theta + mgl\sin\theta - ml^2(\sin\theta\ddot{\theta} + \cos\theta\dot{\theta}^2)\sin\theta = 0$$

$$\ddot{\theta} = g/l\sin\theta$$

Eg. Cartpole



2DOF $q = \begin{bmatrix} x \\ \theta \end{bmatrix}$

$\vec{r}_{1/0} = x\hat{e}_1$ $\quad \vec{v}_{1/0} = \dot{x}\hat{e}_1$ $\quad \vec{a}_{1/0} = \ddot{x}\hat{e}_1$

$\vec{r}_{2/0} = (x + l\sin\theta)\hat{e}_1 - l\cos\theta\hat{e}_2$ $\quad \vec{v}_{2/0} = (\dot{x} + l\cos\theta\dot{\theta})\hat{e}_1 + l\sin\theta\dot{\theta}\hat{e}_2$ $\quad \vec{a}_{2/0} = (\ddot{x} + l\cos\theta\ddot{\theta} - l\sin\theta\dot{\theta}^2)\hat{e}_1 + l(\sin\theta\ddot{\theta} + \cos\theta\dot{\theta}^2)\hat{e}_2$

$Dr_1(q) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\quad Dr_2(q) = \begin{bmatrix} 1 & l\cos\theta \\ 0 & l\sin\theta \end{bmatrix}$

D'Alembert principle: $\sum_{i=1}^{N} (\vec{F}_i^{(act)} - m_i \vec{a}_{i/0})^T Dr_i(q) = 0$

$$\left(\begin{bmatrix} F \\ -Mg \end{bmatrix} - M \begin{bmatrix} \ddot{x} \\ 0 \end{bmatrix} \right)^T \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 0 \\ -mg \end{bmatrix} - m \begin{bmatrix} \ddot{x} + l\cos\theta\ddot{\theta} - l\sin\theta\dot{\theta}^2 \\ l\sin\theta\ddot{\theta} + l\cos\theta\dot{\theta}^2 \end{bmatrix} \right)^T \begin{bmatrix} 1 & l\cos\theta \\ 0 & l\sin\theta \end{bmatrix} = [0, 0]$$

$$\begin{cases} F - M\ddot{x} - m(\ddot{x} + l\cos\theta\ddot{\theta} - l\sin\theta\dot{\theta}^2) = 0 \\ -mgl\sin\theta - m(\ddot{x} + l\cos\theta\ddot{\theta} - l\sin\theta\dot{\theta}^2)l\cos\theta - m(l\sin\theta\ddot{\theta} + l\cos\theta\dot{\theta}^2)l\sin\theta = 0 \end{cases}$$

$$\begin{cases} (M+m)\ddot{x} - F + ml\cos\theta\ddot{\theta} - ml\sin\theta\dot{\theta}^2 = 0 \\ ml\cos\theta\ddot{x} + ml^2\ddot{\theta} + mgl\sin\theta = 0 \end{cases}$$

$$\begin{bmatrix} M+m & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -F - ml\sin\theta\dot{\theta}^2 \\ mgl\sin\theta \end{bmatrix} = 0$$

$M(q)\ddot{q} + F(q, \dot{q}, t) = 0$] D'Alembert always put 2nd-order ODE in this form

mass matrix always invertible

$\ddot{q} = -M(q)F(q, \dot{q}, t)$

the mass matrix is related to kinetic energy $T = \frac{1}{2}\dot{q}M(q)\dot{q}$

· D'Alembert and Kinetic Energy, Lagranges Equation

$\sum_{i=1}^{N} (\vec{F}_i^{(act)} - m_i \vec{a}_{i/0})^T Dr_i(q) = 0$

$\sum_{i=1}^{N} Dr_i(q)^T \vec{F}_i^{(act)} - \sum_{i=1}^{N} m_i Dr_i(q)^T \vec{a}_{i/0} = 0$

Define $Q = \sum_{i=1}^{N} Dr_i(q)^T \vec{F}_i^{(act)} = \sum_{i=1}^{N} \begin{bmatrix} | & | \\ \partial r_i/\partial q_1 & \partial r_i/\partial q_2 \\ | & | \end{bmatrix} \begin{bmatrix} | \\ F_i^{(act)} \\ | \end{bmatrix}$ as generalized force in q direction

kinetic Energy: $T = \sum_{i=1}^{N} \frac{1}{2} m_i \| Dr_i(q)\dot{q} \|_2^2 = \text{function}(q, \dot{q})$

$$\boxed{\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \sum_{i=1}^{N} Dr_i(q)^T \vec{F}_i^{(act)}}$$ Lagrange's Equation

$$\vec{a}_{i/o} = \frac{d}{dt}\vec{V}_{i/o} = \frac{d}{dt}\left(Dr_i(q)\,\dot{q}\right)$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} = \frac{d}{dt}\left[\sum_{i=1}^{N} m_i\, Dr_i(q)^T Dr_i(q)\,\dot{q}\right]$$

$$= \sum_{i=1}^{N} m_i \frac{d}{dt}\left[\nabla r_{i,1}(q) \cdots \nabla r_{i,n}(q)\right] Dr_i(q)\,\dot{q} + \sum_{i=1}^{N} m_i\, Dr_i(q)^T \frac{d}{dt}\left(Dr_i(q)\,\dot{q}\right)$$

$$= \sum_{i=1}^{N} m_i \left[\nabla\dot{r}_{i,1}(q)\,\dot{q} \cdots \nabla\dot{r}_{i,n}(q)\,\dot{q}\right] Dr_i(q)\,\dot{q} + \sum_{i=1}^{N} m_i\, Dr_i(q)^T \vec{a}_{i/o}$$

$$= \sum_{i=1}^{N} m_i \left[\nabla\dot{r}_{i,1}(q)\,\dot{q} \cdots \nabla\dot{r}_{i,n}(q)\,\dot{q}\right] Dr_i(q)\,\dot{q} + \sum_{i=1}^{N} Dr_i(q)^T \vec{F}_i^{(ee)}$$

$$\frac{\partial T}{\partial q} = \frac{\partial}{\partial q}\sum_{i=1}^{N} \frac{1}{2} m_i \left\| Dr_i(q)\,\dot{q} \right\|_2^2$$

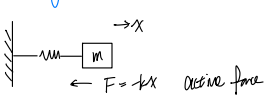
$$= \sum_{i=1}^{N} m_i \frac{\partial}{\partial q}\left(Dr_i(q)\,\dot{q}\right)^T \left(Dr_i(q)\,\dot{q}\right)$$

$$= \sum_{i=1}^{N} m_i \frac{\partial}{\partial q}\left(\begin{bmatrix} \nabla r_{i,1}(q)^T\dot{q} \\ \vdots \\ \nabla r_{i,n}(q)^T\dot{q} \end{bmatrix}\right)^T \left(Dr_i(q)\,\dot{q}\right)$$

$$= \sum_{i=1}^{N} m_i \begin{bmatrix} - & \nabla r_{i,1}(q)^T\dot{q} & - \\ & \vdots & \\ - & \nabla r_{i,n}(q)^T\dot{q} & - \end{bmatrix}^T \left(Dr_i(q)\,\dot{q}\right)$$

$$= \sum_{i=1}^{N} m_i \left[\nabla\dot{r}_{i,1}(q)^T\dot{q} \cdots \nabla\dot{r}_{i,n}(q)^T\dot{q}\right] \left(Dr_i(q)\,\dot{q}\right)$$

Eg. Spring – Mass



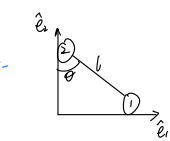
$$T = \frac{1}{2}m\dot{x}^2$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \frac{d}{dt}m\dot{x} = m\ddot{x}$$

$$\sum_{i=1}^{N} Dr_i(q)^T \vec{F}_i^{(ee)} = -kx$$

$$\Big] \; m\ddot{x} = -kx \;\Rightarrow\; F = ma$$

Eg.



1 DOF, $q = \theta$

$$\vec{r}_{1/o} = l\sin\theta\,\hat{e}_1 \qquad \vec{V}_{1/o} = l\cos\theta\,\dot{\theta}\,\hat{e}_1 \qquad Dr_1(q) = l\cos\theta\,\hat{e}_1$$

$$\vec{r}_{2/o} = l\cos\theta\,\hat{e}_2 \qquad \vec{V}_{1/o} = -l\sin\theta\,\dot{\theta}\,\hat{e}_2 \qquad Dr_2(q) = -l\sin\theta\,\hat{e}_2$$

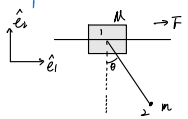
$$T = \frac{1}{2}ml^2\left(\cos^2\theta\,\dot{\theta}^2 + \sin^2\theta\,\dot{\theta}^2\right) = \frac{1}{2}ml^2\dot{\theta}^2$$

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \frac{d}{dt}ml^2\dot{\theta} = ml^2\ddot{\theta}$$

$$\sum_{i=1}^{N} Dr_i(q)^T F_i^{(ee)} = \begin{bmatrix} l\cos\theta \\ 0 \end{bmatrix}^T \begin{bmatrix} 0 \\ -mg \end{bmatrix} + \begin{bmatrix} 0 \\ -l\sin\theta \end{bmatrix}^T \begin{bmatrix} 0 \\ -mg \end{bmatrix} = mgl\sin\theta$$

$$\Big] \; \ddot{\theta} = \frac{g}{l}\sin\theta$$

Eg Cartpole



2 DoF $q = \begin{bmatrix} x \\ \theta \end{bmatrix}$

$\vec{r}_{1/0} = x\hat{e}_1$ $\quad$ $\vec{v}_{1/0} = \dot{x}\hat{e}_1$

$\vec{r}_{2/0} = (x + l\sin\theta)\hat{e}_1 - l\cos\theta\hat{e}_2$ $\quad$ $\vec{v}_{2/0} = (\dot{x} + l\cos\theta\dot{\theta})\hat{e}_1 + l\sin\theta\dot{\theta}\hat{e}_2$

$Dr_1(q) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\quad$ $Dr_2(q) = \begin{bmatrix} 1 & l\cos\theta \\ 0 & l\sin\theta \end{bmatrix}$

$T = \frac{1}{2}M\|\dot{x}\hat{e}_1\|^2 + \frac{1}{2}m\|(\dot{x} + l\cos\theta\dot{\theta})\hat{e}_1 + l\sin\theta\dot{\theta}\hat{e}_2\|^2$

$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x} + l\cos\theta\dot{\theta})^2 + \frac{1}{2}m(l\sin\theta\dot{\theta})^2$

$= \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\cos\theta\dot{\theta}\dot{x} = \frac{1}{2}\begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}^T \begin{bmatrix} M+m & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \frac{d}{dt}\left(\begin{bmatrix} (M+m)\dot{x} + ml\cos\theta\dot{\theta} \\ ml^2\dot{\theta} + ml\cos\theta\dot{x} \end{bmatrix}\right) - \begin{bmatrix} 0 \\ ml\sin\theta\dot{\theta}\dot{x} \end{bmatrix}$

$= \begin{bmatrix} (M+m)\ddot{x} + ml\cos\theta\ddot{\theta} - ml\sin\theta\dot{\theta}^2 \\ ml^2\ddot{\theta} + ml\cos\theta\ddot{x} \end{bmatrix}$

$\sum_i Dr_i(q)^T F_i^{(ext)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} F \\ -Mg \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ l\cos\theta & l\sin\theta \end{bmatrix}\begin{bmatrix} 0 \\ -mg \end{bmatrix} = \begin{bmatrix} F \\ -mgl\sin\theta \end{bmatrix}$

$\begin{bmatrix} (M+m) & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix}\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F + ml\sin\theta\dot{\theta}^2 \\ -mgl\sin\theta \end{bmatrix}$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad$ $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} \dot{p} \\ \ddot{\theta} \end{bmatrix} = \frac{1}{(M+m)ml^2 - m^2l^2\cos^2\theta}\begin{bmatrix} ml^2 & -ml\cos\theta \\ -ml\cos\theta & M+m \end{bmatrix}\begin{bmatrix} F + ml\sin\theta\dot{\theta}^2 \\ -mgl\sin\theta \end{bmatrix}$

$= \frac{1}{Mml^2 + m^2l^2\sin^2\theta}\begin{bmatrix} ml^2F + m^2l^3\sin\theta\dot{\theta}^2 + m^2gl^2\sin\theta\cos\theta \\ -ml\cos\theta F - m^2l^2\sin\theta\cos\theta\dot{\theta}^2 - (M+m)mgl\sin\theta \end{bmatrix}$

$= \begin{bmatrix} \frac{F + ml\sin\theta\dot{\theta}^2 + mg\sin\theta\cos\theta}{M + m\sin^2\theta} \\ \frac{-\cos\theta F - ml\sin\theta\cos\theta\dot{\theta}^2 - (M+m)g\sin\theta}{l(M + m\sin^2\theta)} \end{bmatrix}$

· Lagrange's Equation & conservative Force

active force on particle i: $F_i^{(ext)} = F_{i,c}^{(ext)} + F_{i,nc}^{(ext)}$

$F_{i,c}^{(ext)}$: conservative force, from potential energy

$F_{i,nc}^{(ext)}$: friction, control force ... don't know if it's from a potential energy, treat it as non-conservative force

total potential energy

$U(r_1 \cdots r_N) = U(r_1(q) \cdots r_N(q))$

$F_{i,c}^{(ext)} = -\nabla_{r_i} U(r_1(q) \cdots r_N(q))$

$Q = \sum_{i=1}^N Dr_i(q)^T F_i^{(ext)}$

$= \sum_{i=1}^N Dr_i(q)^T F_{i,c}^{(ext)} + Dr_i(q)^T F_{i,nc}^{(ext)}$

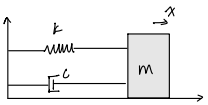
$= \underbrace{-\nabla_q U(r_1(q) \cdots r_N(q))}_{\text{conservative}} + \underbrace{\sum_{i=1}^N Dr_i(q)^T F_{i,nc}^{(ext)}}_{\text{non-conservative}}$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \sum_{i=1}^N Dr_i(q)^T F_i^{(ext)} = -\nabla_q U(r_1(q) \cdots r_N(q)) + \sum_{i=1}^N Dr_i(q)^T F_{i,nc}^{(ext)}$

define the Lagrangian $L = T - U$ = kinetic energy − potential energy

$$\boxed{\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \sum_{i=1}^N Dr_i(q)^T F_{i,nc}^{(ext)}}$$

Eg. Spring – Mass – Damper



$U_1 + T_1 = U_2 + T_2$

$U_1 - U_2 = T_2 - T_1 = \int_{x_1}^{x_2} -kx\,dx = \frac{k}{2}x_1^2 - \frac{k}{2}x_2^2$

1 DoF $q = x$

$T = \frac{1}{2}m\dot{x}^2 \quad U = \frac{k}{2}x^2$

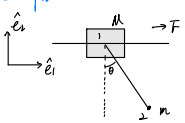
$L = T - U = \frac{1}{2}m\dot{x}^2 - \frac{k}{2}x^2$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = m\ddot{x} + kx$

$\sum_{i=1}^{n} Dr_i(q)^T F_{i,nc}^{(act)} = -c\dot{x}$

$\Big\} \quad m\ddot{x} + kx = -c\dot{x}$

Eg. Cartpole



2 DoF $q = \begin{bmatrix} x \\ \theta \end{bmatrix}$

$\vec{r}_{1/0} = x\hat{e}_1 \qquad \vec{v}_{1/0} = \dot{x}\hat{e}_1$

$\vec{r}_{2/0} = (x + l\sin\theta)\hat{e}_1 - l\cos\theta\,\hat{e}_2 \qquad \vec{v}_{2/0} = (\dot{x} + l\cos\theta\dot{\theta})\hat{e}_1 + l\sin\theta\dot{\theta}\,\hat{e}_2$

$Dr_1(q) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad Dr_2(q) = \begin{bmatrix} 1 & l\cos\theta \\ 0 & l\sin\theta \end{bmatrix}$

$T = \frac{1}{2}M\|\dot{x}\hat{e}_1\|_2^2 + \frac{1}{2}m\|(\dot{x} + l\cos\theta\dot{\theta})\hat{e}_1 + l\sin\theta\dot{\theta}\hat{e}_2\|_2^2$

$= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x} + l\cos\theta\dot{\theta})^2 + \frac{1}{2}m(l\sin\theta\dot{\theta})^2$

$= \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\cos\theta\dot{\theta}\dot{x} = \frac{1}{2}\begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}^T \begin{bmatrix} M+m & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}$

$U = -mgl\cos\theta$

$L = T - U = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\cos\theta\dot{\theta}\dot{x} + mgl\cos\theta$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \frac{d}{dt}\left(\begin{bmatrix} (M+m)\dot{x} + ml\cos\theta\dot{\theta} \\ ml^2\dot{\theta} + ml\cos\theta\dot{x} \end{bmatrix}\right) - \begin{bmatrix} 0 \\ -ml\sin\theta\dot{\theta}\dot{x} - mgl\sin\theta \end{bmatrix}$

$= \begin{bmatrix} (M+m)\ddot{x} + ml\cos\theta\ddot{\theta} - ml\sin\theta\dot{\theta}^2 \\ ml^2\ddot{\theta} + ml\cos\theta\ddot{x} - ml\sin\theta\dot{\theta}\dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ ml\sin\theta\dot{\theta}\dot{x} + mgl\sin\theta \end{bmatrix}$

$\sum_{i=1}^{n} Dr_i(q)^T F_{i,nc}^{(act)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} F \\ 0 \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$

$\Big\} \quad \begin{bmatrix} M+m & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix}\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F + ml\sin\theta\dot{\theta}^2 \\ -mgl\sin\theta \end{bmatrix}$

Canonical Form

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \sum_{i=1}^{n} Dr_i(q)^T F_{i,nc}^{(ext)} \qquad L = T - U = \frac{1}{2}\dot{q} M(q) \dot{q} - U(q)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \frac{d}{dt}(M(q)\dot{q}) - \frac{\partial}{\partial q}\left[\frac{1}{2}\dot{q} M(q)\dot{q} - U(q)\right]$$

$$= M(q)\ddot{q} + \underbrace{\left[\dot{M}(q)\dot{q} - \frac{\partial}{\partial q}\frac{1}{2}\dot{q} M(q)\dot{q}\right]}_{C(q,\dot{q}):\ \text{Coriolis term}} + \underbrace{\frac{\partial U}{\partial q}}_{G(q):\ \text{potential term}} = \sum_{i=1}^{n} Dr_i(q)^T F_{i,nc}^{(ext)}$$

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \sum_{i=1}^{n} Dr_i(q)^T F_{i,nc}^{(ext)}$$

Manipulator Equation